In Eq. 4.30, the factors α_j (0) are the values of the coefficients at time zero. If we know the state-vector |Ψ> at time zero, then the coefficients are given by the projections of |Ψ> on the basis eigenvectors. We can write this as

$$\alpha_j(0) = \langle E_j|\Psi(0)\rangle. \qquad (4.31)$$

Now let's put the whole thing together and write the full solution of the time-dependent Schrödinger equation:

$$|\Psi(t)\rangle = \sum_j \alpha_j(0)\ e^{-\frac{i}{\hbar}E_j t}\ |E_j\rangle.$$

When we use Eq. 4.31 to replace α_j(0), this equation becomes

$$|\Psi(t)\rangle = \sum_j \langle E_j|\Psi(0)\rangle\ e^{-\frac{i}{\hbar}E_j t}\ |E_j\rangle. \qquad (4.32)$$

Eq. 4.32 can be written in the more elegant form,

$$|\Psi(t)\rangle = \sum_j |E_j\rangle\langle E_j|\Psi(0)\rangle\ e^{-\frac{i}{\hbar}E_j t}, \qquad (4.33)$$

which emphasizes that we're summing over the basis vectors. You may wonder how we just happen to "know" |Ψ(0)>. The answer depends on the circumstances, but usually, we assume we can use some apparatus to prepare the system in a known state.

Before we discuss the bigger meaning of these equations, I want to restate them as a recipe. I'll assume you already know enough about the system and its space of states to get started.